where C is again a constant.

Finally, another term, a result of the electrical repulsion of the protons, must be added. Here we are in the familiar territory of electrostatics. The charge on a nucleus is Ze, where e is the elementary quantum of electricity. The electric energy of a nucleus is proportional to the square of its charge, i.e. to Ze^2 just as the energy of a condenser is proportional to the square of its charge. Furthermore it is inversely proportional to the radius, r, of the nucleus. There is also a numerical factor to be considered, which for a homogeneously charged sphere is 3/5 If the charge is displaced slightly toward the surface, this factor becomes smaller and will approach $\frac{1}{2}$. Since the radius is known, anyway, not very exactly, we can retain the factor 3/5spite the fact that there is undoubtedly a certain displacement of the charge toward the surface. This energy component may therefore be written:

$$E_c = \frac{3}{5}\frac{(Ze)^2}{r}$$

But since the radius r is proportional to the cube root of the volume, i.e. to the cube root of the number of particles, we can write: $r = r_0 (N + Z)^{1/3}$, where r_0 is a constant which would, so to speak, be the radius of a nucleus containing one single particle, but naturally must not be identified with the radius of the proton or the neutron. Thus, this portion of the binding energy per individual particle is:

$$\frac{E_c}{N+Z} = \frac{3}{5}\frac{(Ze)^2}{(N+Z)^{4/3} r_0}$$

We must also add this term to our equation with a positive sign, since the electric repulsion reduces the quantity of the total binding energy. The complete expression of the binding energy per particle, finally, is:

$$\frac{E}{N+Z} = -A + B\frac{(N-Z)^2}{(N+Z)^2} + \frac{C}{(N+Z)^{1/3}} + \frac{3}{5}\frac{(Ze)^2}{(N+Z)^{4/3} r_0}$$

This equation contains four constants, A, B, C and r_0, of which, to start with, we know only r_0—and even this one rather inexactly—from the measurements of nuclei.

In order to make use of this equation, it is essential to know the four